MTH107 2023S1 Exam

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ENG301 Microprocessor Programming
* ENG355 Machine Learning
* ICT133 Structured Programming
* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
* MTH320 Mathematical Methods II
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH107

Examination - January Semester 2023

Calculus I

Tuesday, 16 May 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

MTH107 Copyright © 2023 Singapore University of Social Sciences (SUSS)

Answer all questions. (Total 100 marks)

Question 1a

Determine

$$\lim_{x \to \sqrt{5}} \frac{x^2 - 5}{x - \sqrt{5}}$$

without using L'Hopital's Rule. Show your steps clearly.

(6 marks)

Question 1b

Determine if the following limit exists.

$$\lim_{x \to -5} \frac{x + 2}{x + 5}$$

Show your steps clearly.

(7 marks)

Question 1c

Show that there exists a positive real root for the equation

$$x^4 = x^2 + 1.$$

Justify your answer.

(7 marks)

Question 2a

Let f be the function defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} (x-1)^3 \cos(\frac{1}{1-x}) & \text{if } x \neq 1, \\ 0 & \text{if } x = 1. \end{cases}$$

Discuss whether f is continuous at $x = 1$.

(7 marks)

Question 2b

Let f be the function defined by

$$f(x) = \begin{cases} \frac{x^2-a}{x-2} & \text{if } x \neq 2, \\ b & \text{if } x = 2. \end{cases}$$

Find all possible values of $a, b \in \mathbb{R}$ such that f is continuous at $x = 2$. Justify your answer.

(7 marks)

Question 2c

Let

$$f(x) = |x| + 2x.$$

Is f differentiable at 0? If yes, determine the value of $f'(0)$. Justify your answer.

(6 marks)

Question 3a

Determine

$$\lim_{x \to 0} \left(\frac{2023}{x} - \frac{2023}{\sin x} \right).$$

Show your steps clearly.

(8 marks)

Question 3b

Let f be continuous and differentiable on $[-8, 0]$. Suppose that $f(-8) = 6$ and

$$f'(x) \leq 3$$

for all x. What is the largest possible value for $f(0)$?

(12 marks)

Question 4a

Determine the Taylor series for x^4 about $a = 2$.

(10 marks)

Question 4b

Determine the Taylor expansion of

$$\ln(x + e^2)$$

about $x = 0$, up to and including the term in x^3. Use this Taylor expansion to show that

$$\ln(e^{-2} + e^2) \approx 2 + \frac{a}{e^4} + \frac{b}{e^8} + \frac{c}{e^{12}}$$

where $a, b, c \in \mathbb{R}$ are constants to be determined.

(10 marks)

Question 5

Define the function $f : \mathbb{R}\backslash\{\pm 3\} \longrightarrow \mathbb{R}$ by

$$f(x) = \frac{2x^2 + x - 9}{9 - x^2}.$$

Question 5a

Apply the first derivative test to compute the intervals on which f is increasing.

(6 marks)

Question 5b

Determine the horizontal and vertical asymptotes of f if they exist. Justify your answers.

(7 marks)

Question 5c

Determine the critical point(s) of f. Apply relevant derivative tests and discuss the nature of these critical point(s). Show your steps clearly.

(7 marks)

----- END OF PAPER -----

MTH107 Copyright © 2023 Singapore University of Social Sciences (SUSS)

1(a) $\lim_{x \to \sqrt{5}} \frac{x^2-5}{x-\sqrt{5}} = \lim_{x \to \sqrt{5}} \frac{x^2-5}{x-\sqrt{5}}$

$= \lim_{x \to \sqrt{5}} \frac{(x-\sqrt{5})(x+\sqrt{5})}{x-\sqrt{5}}$

$= \lim_{x \to \sqrt{5}} (x+\sqrt{5})$

$= 2\sqrt{5}$

(b) $\lim_{x \to -5^+} \frac{x+2}{x+5} = \lim_{x \to -5^+} \frac{(x+5)-3}{x+5}$

$= \lim_{x \to -5^+} \left(1 - \frac{3}{x+5}\right)$

$= 1 - \lim_{x \to -5^+} \frac{3}{x+5}$

$= -\infty$

$\lim_{x \to -5^-} \frac{x+2}{x+5} = \lim_{x \to -5^-} \left(1 - \frac{3}{x+5}\right)$

$= 1 - \lim_{x \to -5^-} \frac{3}{x+5}$

$= 1 - (-\infty)$

$= \infty$

$\therefore$ The limit doesn't exist

(c) $x^4 = x^2 + 1$

$x^4 - x^2 - 1 = 0$

let $f(x) = x^4 - x^2 - 1$

$f(1) = 1 - 1 - 1 = -1$

$f(2) = 16 - 4 - 1 = 13$

$f(1) < 0$, $f(2) > 0$

By Intermediate Value Theorem,

$\exists c$ on $(1,2)$ such that $f(c) = 0$

$\therefore$ There exists a positive real root

2(a)

$$-|(x-1)^3| \le -|(x-1)^3||\cos\tfrac{1}{1-x}| \le (x-1)^3\cos\tfrac{1}{1-x} \le |(x-1)^3||\cos\tfrac{1}{1-x}| \le |(x-1)^3|$$

$$-|(x-1)^3| \le (x-1)^3\cos\tfrac{1}{1-x} \le |(x-1)^3|$$

$$\lim_{x\to1} -|(x-1)^3| \le \lim_{x\to1}(x-1)^3\cos\tfrac{1}{1-x} \le \lim_{x\to1}|(x-1)^3|$$

$$0 \le \lim_{x\to1}(x-1)^3\cos\tfrac{1}{1-x} \le 0$$

By Squeeze Theorem,

$$\lim_{x\to1}(x-1)^3\cos\tfrac{1}{1-x} = 0$$

$$\lim_{x\to1} f(x) = 0$$

since $\lim_{x\to1} f(x) = f(1) = 0$

$\therefore$ $f(x)$ is continuous at $x=1$

(b) let $x^2-a = (x-2)(x+c)$

$x^2-a = x^2+(c-2)x-2c$

$\Leftrightarrow$ $c-2=0$, $a=2c$

$c=2$

$a=2c=4$

$$\lim_{x\to2} f(x) = \lim_{x\to2}\frac{x^2-a}{x-2} = \lim_{x\to2}\frac{(x-2)(x+2)}{x-2} = \lim_{x\to2}(x+2) = 4$$

since $f(x)$ is continuous,

$$b = f(0) = \lim_{x\to2} f(x) = 4$$

$\therefore$ $a = b = 4$

2(c) $f(0) = 0$

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} \left[|x| + 2x \right] = 0$$

$\therefore f(x)$ is continuous at $x = 0$

$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \frac{h + 2h}{h} = 3$$

$$\lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = \frac{-h + 2h}{h} = 1$$

$$\lim_{h \to 0} \frac{f(h) - f(0)}{h} \text{ doesn't exist}$$

$\therefore f(x)$ is not differentiable

3(a)
$$\lim_{x \to 0} \left(\frac{2023}{x} - \frac{2023}{\sin x} \right) = 2023 \lim_{x \to 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$$

$$= 2023 \lim_{x \to 0} \frac{\sin x - x}{x \cdot \sin x}$$

$\frac{0}{0} \to$ L'Hopital Rule

$$= 2023 \lim_{x \to 0} \frac{\cos x - 1}{\sin x + x \cos x}$$

$\frac{0}{0} \to$ L'Hopital Rule

$$= 2023 \lim_{x \to 0} \frac{-\sin x}{2\cos x - x \sin x}$$

$$= 2023 \cdot \frac{-\sin 0}{2\cos 0 - 0}$$

$$= 2023 \cdot \frac{0}{2}$$

$$= 0$$

3(b) By Mean Value Theorem,

$\exists c$ on $(-8, 0)$ such that $f'(c) = \frac{f(0) - f(-8)}{0 - (-8)} = \frac{f(0) - 6}{8}$

$$f'(x) \leq 3$$

$$\Rightarrow f'(c) \leq 3$$

$$\frac{f(0) - 6}{8} \leq 3$$

$$f(0) - 6 \leq 24$$

$$f(0) \leq 30$$

$\therefore \max f(0) = 30$

4(a) let $f(x) = x^4$ $\qquad f(2) = 16$

$f^{(1)}(x) = 4x^3$ $\qquad f^{(1)}(2) = 32$

$f^{(2)}(x) = 12x^2$ $\qquad f^{(2)}(2) = 48$

$f^{(3)}(x) = 24x$ $\qquad f^{(3)}(2) = 48$

$f^{(4)}(x) = 24$ $\qquad f^{(4)}(2) = 24$

$$x^4 = \sum_{n=0}^{4} \frac{f^{(n)}(a)}{n!}(x-a)^n \quad \text{where } a = 2$$

$$= 16 + 32(x-2) + 24(x-2)^2 + 8(x-2)^3 + (x-2)^4$$

(b) $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$

$$\ln(x+e^2) = \ln e^2\left(1+\frac{x}{e^2}\right)$$

$$= \ln e^2 + \ln\left(1+\frac{x}{e^2}\right)$$

$$= 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\left(\frac{x}{e^2}\right)^n}{n}$$

$$= 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(xe^{-2})^n}{n}$$

$$= 2 + \frac{x}{e^2} - \frac{x^2}{2e^4} + \frac{x^3}{3e^6} + \dots$$

$$\ln(e^{-2}+e^2) \simeq 2 + \frac{e^{-2}}{e^2} - \frac{e^{-4}}{2e^8} + \frac{e^{-6}}{3e^6}$$

$$= 2 + \frac{1}{e^4} - \frac{\frac{1}{2}}{e^8} + \frac{\frac{1}{3}}{e^{12}}$$

$\therefore a = 1$

$\therefore b = -\frac{1}{2}$

$\therefore c = \frac{1}{3}$

5(a) $f(x) = \dfrac{2x^2+x-9}{9-x^2}$

$$f'(x) = (4x+1)(9-x^2)^{-1} + (2x^2+x-9)(-1)(9-x^2)^{-2}(-2x)$$

$$= \frac{(4x+1)(9-x^2) + 2x(2x^2+x-9)}{(9-x^2)^2}$$

$$= \frac{x^2+18x+9}{(9-x^2)^2}$$

let $f'(x) > 0$

$\Rightarrow x^2+18x+9 > 0$

$(x+9)^2 - 72 > 0$

$(x+9)^2 > 72$

$x+9 > \sqrt{72}$ or $x+9 < -\sqrt{72}$

$x+9 > 6\sqrt{2}$ or $x+9 < -6\sqrt{2}$

$x > 6\sqrt{2}-9$ or $x < -9-6\sqrt{2}$

$\therefore$ $f(x)$ is increasing when $x > 6\sqrt{2}-9$ or $x < -9-6\sqrt{2}$

(b) $f(x) = \dfrac{2x^2+x-9}{9-x^2}$

$$= -\frac{2x^2+x-9}{x^2-9}$$

$$= -\frac{2(x^2-9)+x+9}{x^2-9}$$

$$= -2 - \frac{x+9}{(x+3)(x-3)}$$

$$= -2 + \frac{1}{x+3} - \frac{2}{x-3}$$

$\cancel{\lim_{x\to 3} f(x) = \lim_{x\to 3}\left[-2+\frac{1}{x+3}-\frac{2}{x-3}\right] = -\infty}$

$\cancel{\lim_{x\to -3} f(x) = \lim_{x\to -3}\left[-2+\frac{1}{x+3}-\frac{2}{x-3}\right] = -\infty}$

$\therefore$ The vertical asymptotes are $x=3$ and $x=-3$

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left[-2+\frac{1}{x+3}-\frac{2}{x-3}\right] = -2$

$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty}\left[-2+\frac{1}{x+3}-\frac{2}{x-3}\right] = -2$

$\therefore$ The horizontal asymptote is $y=-2$

5(c) $f(x) = -2 + \frac{1}{x+3} - \frac{2}{x-3}$

$$f'(x) = \frac{2}{(x-3)^2} - \frac{1}{(x+3)^2}$$

$$f''(x) = \frac{2}{(x+3)^3} - \frac{4}{(x-3)^3}$$

let $f'(x) = 0$

$$\Rightarrow \frac{2}{(x-3)^2} = \frac{1}{(x+3)^2}$$

$$2(x+3)^2 = (x-3)^2$$

$$2(x^2+6x+9) = x^2-6x+9$$

$$x^2+18x+9 = 0$$

$$(x+9)^2 - 72 = 0$$

$$(x+9)^2 = 72$$

$$x+9 = \pm\sqrt{72}$$

$$x = -9 \pm \sqrt{72}$$

$f(-9-\sqrt{72}) = -1.971$

$f''(-9-\sqrt{72}) = -0.000193$

$(-9-\sqrt{72}, -1.971)$ is a local maxima

$f(-9+\sqrt{72}) = -1.029$

$f''(-9+\sqrt{72}) = 0.2224$

$(-9+\sqrt{72}, -1.029)$ is a local minima